5-5 INFERENCE ON THE RATIO OF VARIANCES OF TWO NORMAL POPULATIONS

- **Scope**: The procedures are designed to evaluate and compare the variances of two distinct populations as shown in Fig. 5-1.

- **Core Assumption**: Both populations are assumed to be normally distributed.

- **Sensitivity**: Both hypothesis testing and confidence interval procedures are relatively **sensitive to the normality assumption**, meaning the results may be unreliable if the populations are not normal.

5-5.1 Hypothesis Testing on the Ratio of Two Variances

5-5.1 Hypothesis Testing on the Ratio of Two Variances

- **Scenario**: Two independent normal populations are compared where the means (μ_1, μ_2) and variances (σ_1^2, σ_2^2) are unknown.

- **Hypothesis**: The goal is to test the equality of the two variances, specifically H_0: $\sigma_1^2 = \sigma_2^2$ against H_1: $\sigma_1^2 \neq \sigma_2^2$.

- **Data Source**: Random samples of size n_1 and n_2 are taken from the respective populations, providing sample variances s_1^2 and s_2^2.

- **Requirement**: A **new probability distribution** is necessary to develop the formal test procedure for these hypotheses.

- **The F Distribution**: This is recognized as one of the most essential probability distributions in the field of statistics.

- **Definition**: The random variable F is established as the ratio of two independent chi-square random variables, where each is divided by its respective degrees of freedom.

- **Mathematical Form**: It is expressed as $F = (W/u)/(Y/v)$.

- **Components**: In this expression, W and Y are independent chi-square variables with u and v degrees of freedom, respectively.

- **Definition**: **The F Distribution** is constructed from the ratio of two independent chi-square random variables, W and Y, with u and υ degrees of freedom, respectively. The random variable is defined as:

$$F = \frac{W/u}{Y/\upsilon} \qquad (5-18)$$

5-5.1 Hypothesis Testing on the Ratio of Two Variances

- **Probability Density Function**: The distribution of this ratio follows a specific density function, $f(x)$:

$$f(x) = \frac{\Gamma(\frac{u+v}{2})(\frac{u}{v})^{u/2} x^{u/2-1}}{\Gamma(\frac{u}{2})\Gamma(\frac{v}{2})(\frac{ux}{v}+1)^{(u+v)/2}}, 0 < x < \infty, \qquad \textbf{(5-19)}$$

- **Notation**: A random variable following this distribution is said to have u degrees of freedom in the numerator and v degrees of freedom in the denominator, commonly abbreviated as $F_{u,v}$.

- The key characteristics of the F distribution are summarized as follows:

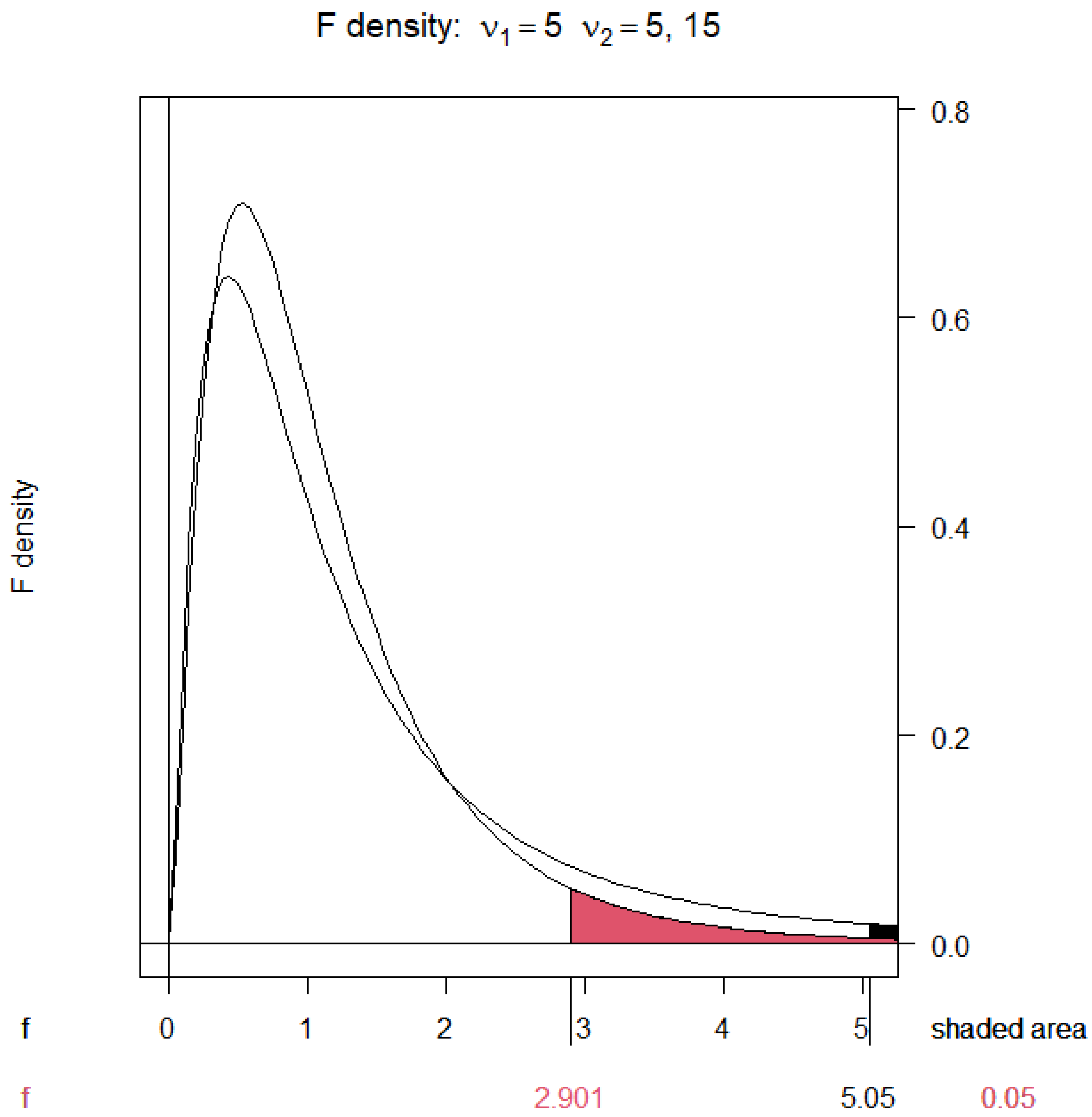
  - **Statistical Parameters**: The mean is defined as $\mu = \upsilon/(\upsilon - 2)$ for $\upsilon > 2$, and the variance is given by $\sigma^2 = 2\upsilon^2(u + \upsilon - 2)/[u(\upsilon - 2)^2(\upsilon - 4)]$ for $\upsilon > 4$.

  - **Distribution Shape**: F random variables are non-negative, and the density functions are skewed to the right.

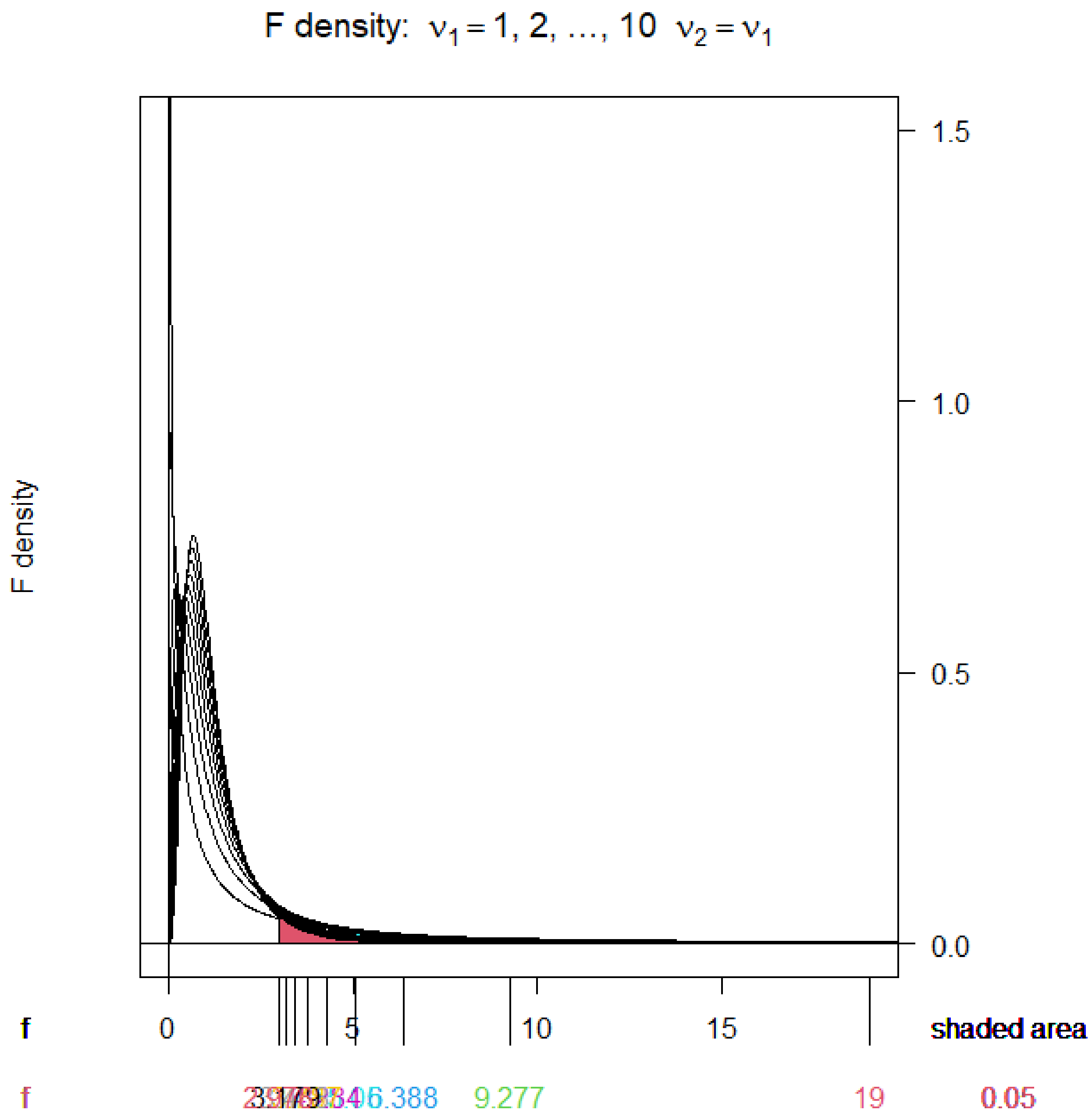
  - **Comparison and Flexibility**: As shown in **Fig. 5-4**, while the F distribution resembles the chi-square distribution, the two parameters (u and υ) provide extra flexibility in determining the exact shape.

- **Visual Representation**: **Fig. 5-4** illustrates the probability density functions for two different F distributions, highlighting how the curves change based on the degrees of freedom.

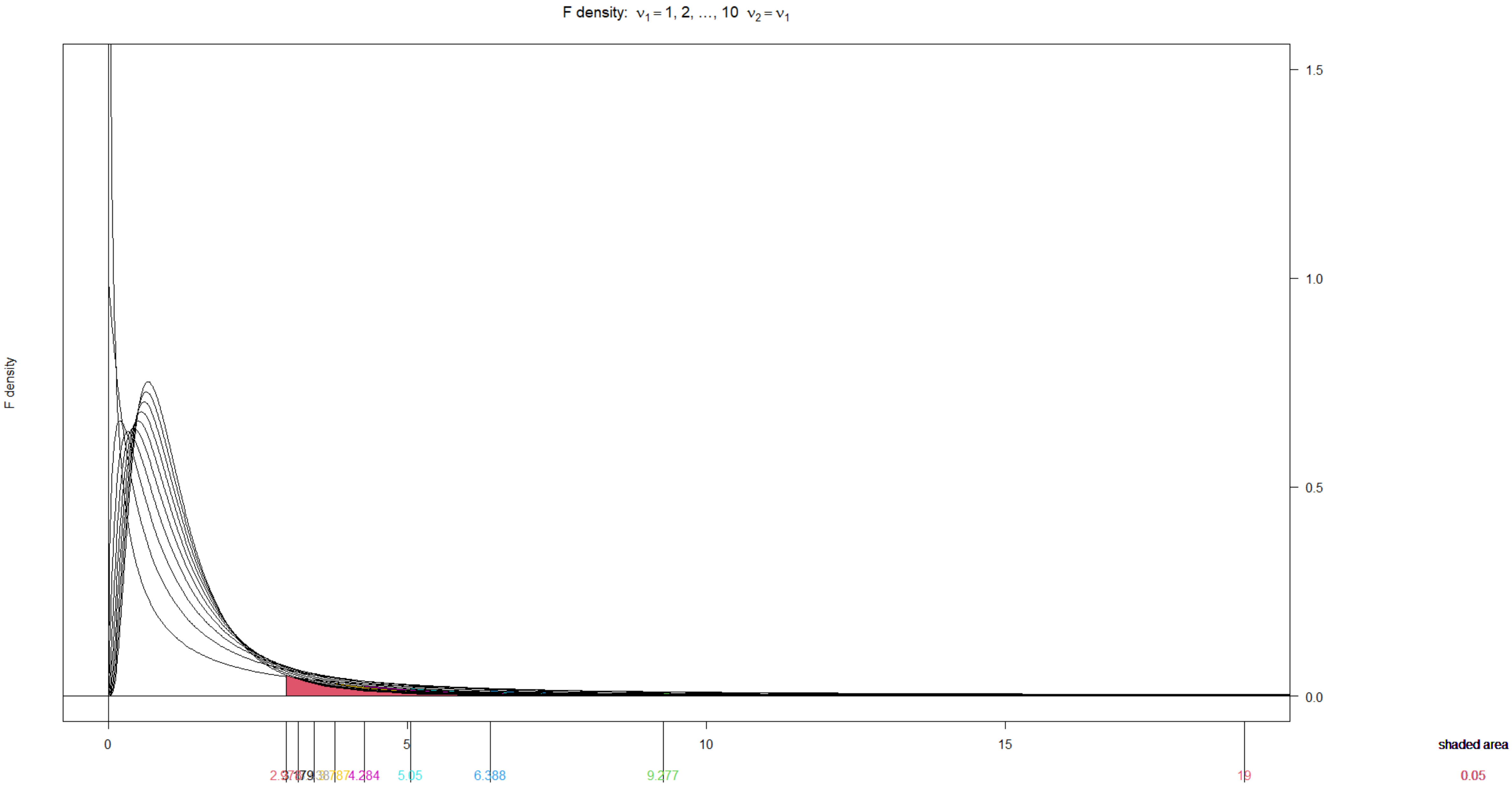
5-5.1 Hypothesis Testing on the Ratio of Two Variances



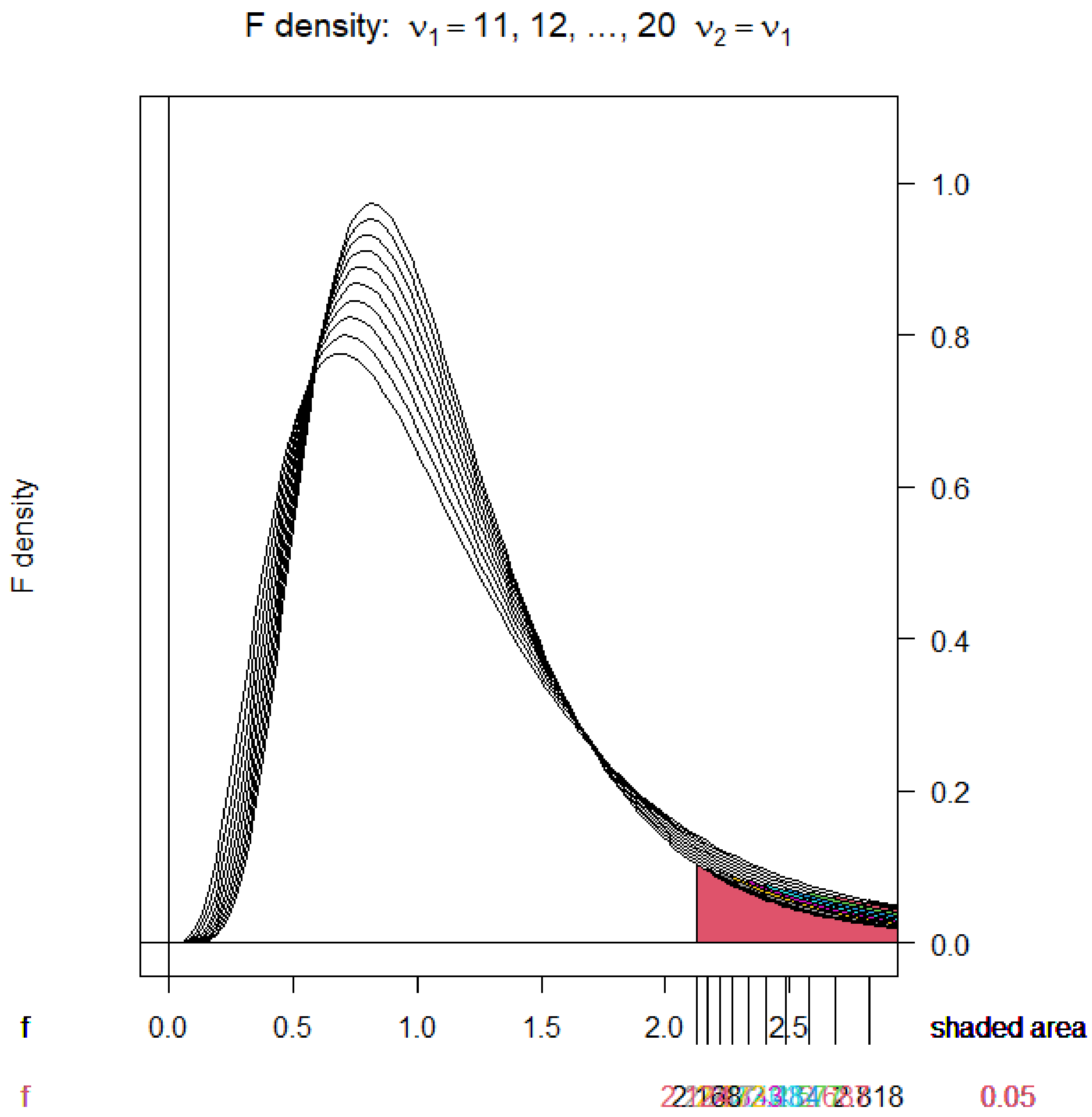
5-5.1 Hypothesis Testing on the Ratio of Two Variances



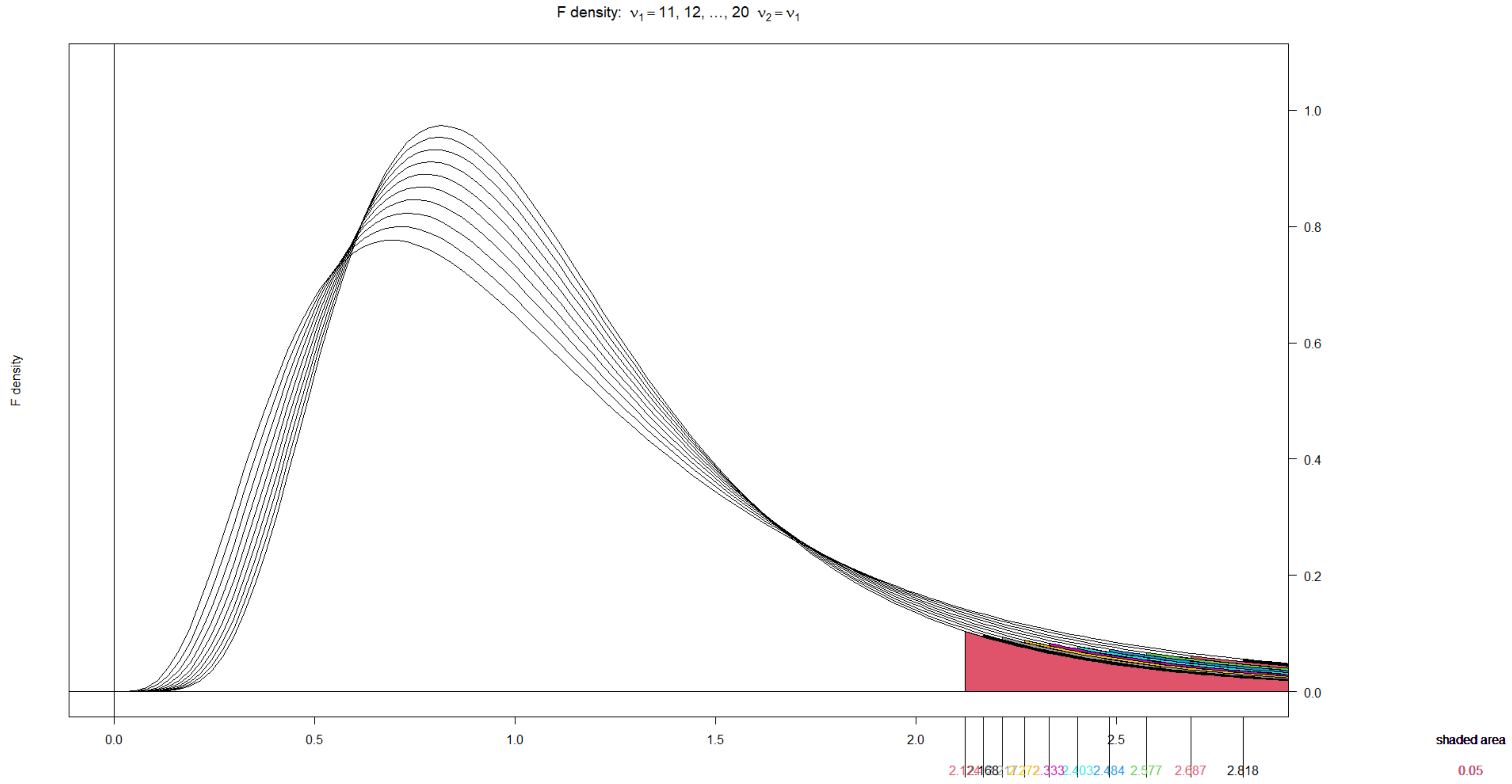
5-5.1 Hypothesis Testing on the Ratio of Two Variances



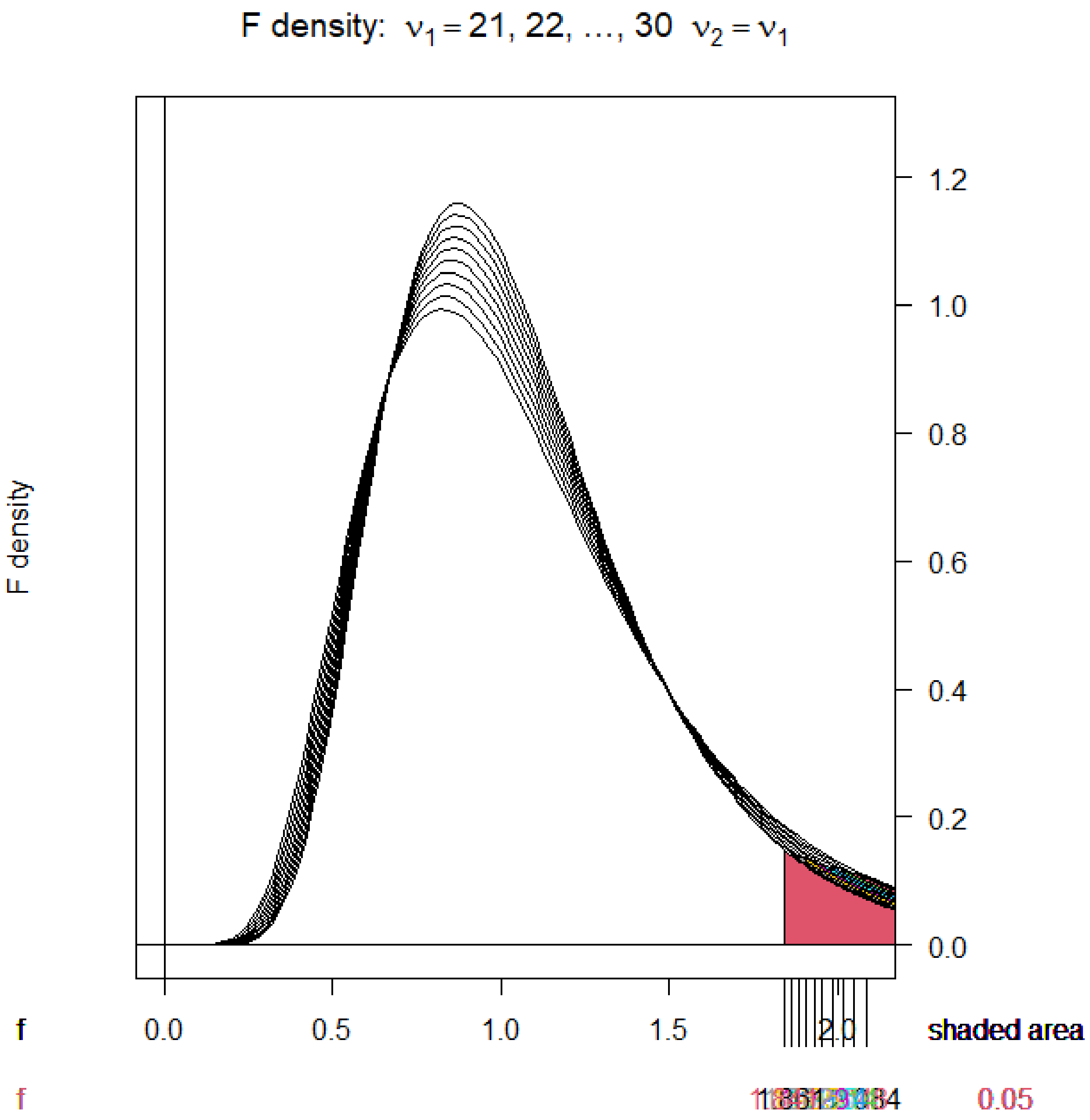
5-5.1 Hypothesis Testing on the Ratio of Two Variances



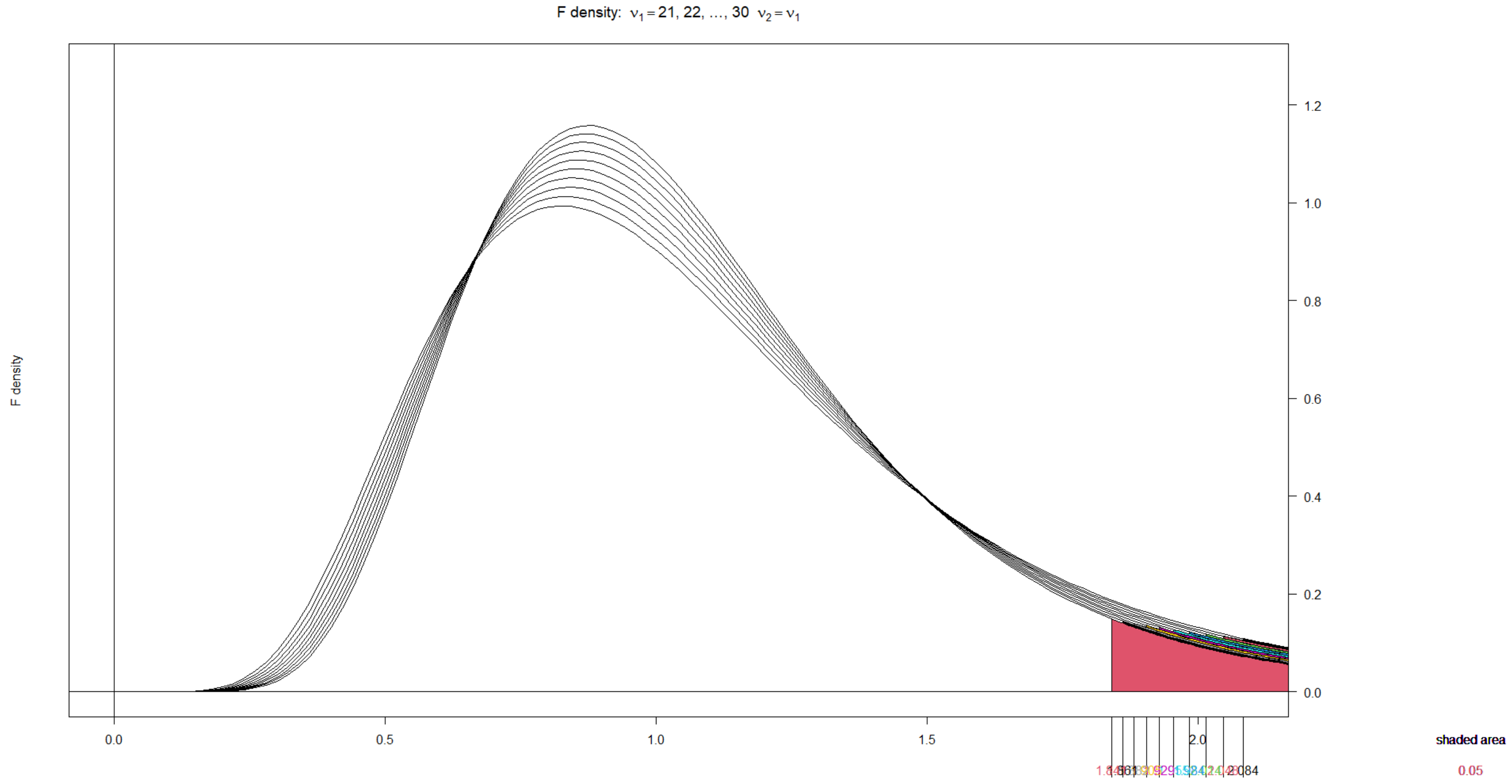
5-5.1 Hypothesis Testing on the Ratio of Two Variances



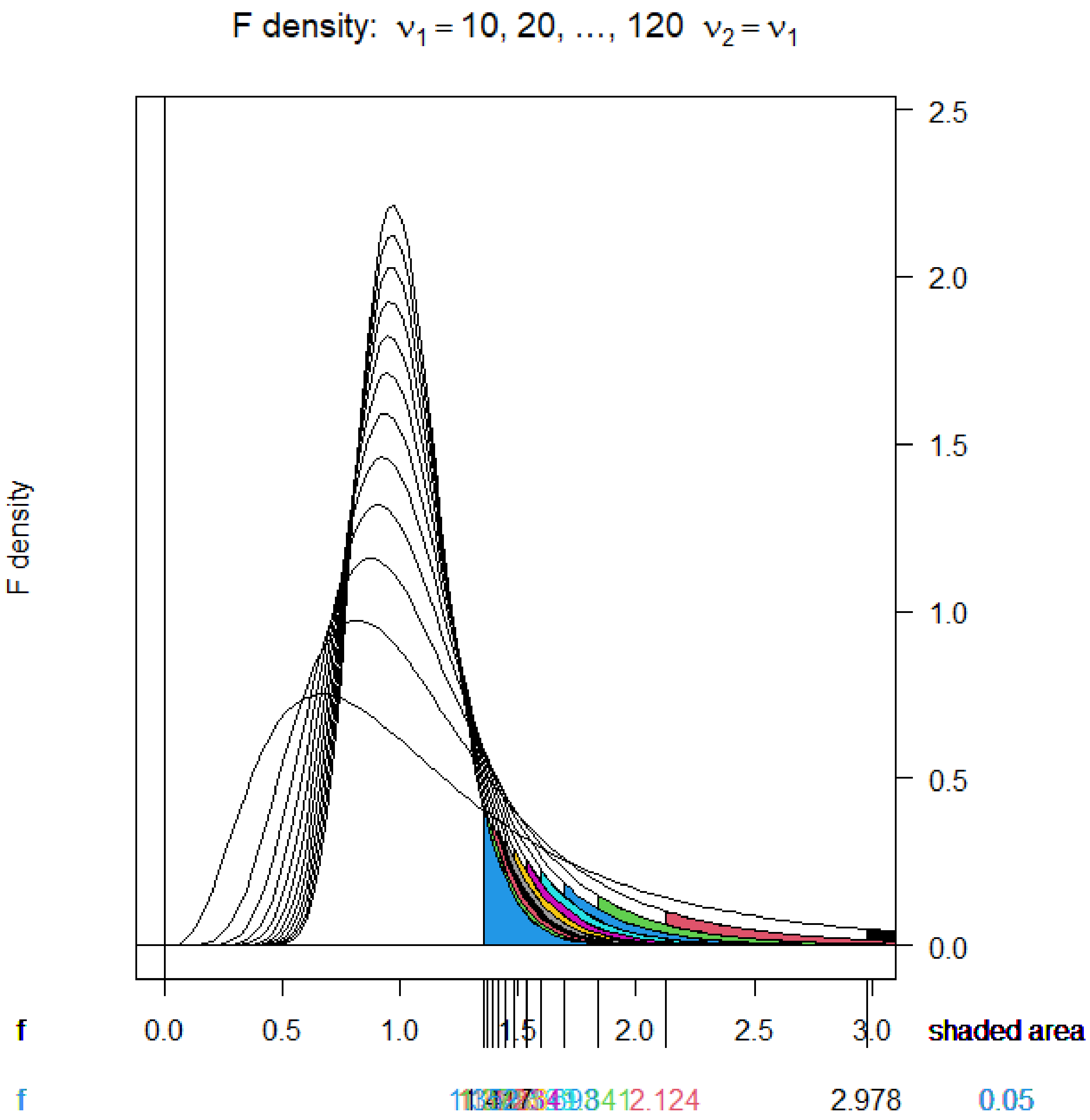
5-5.1 Hypothesis Testing on the Ratio of Two Variances



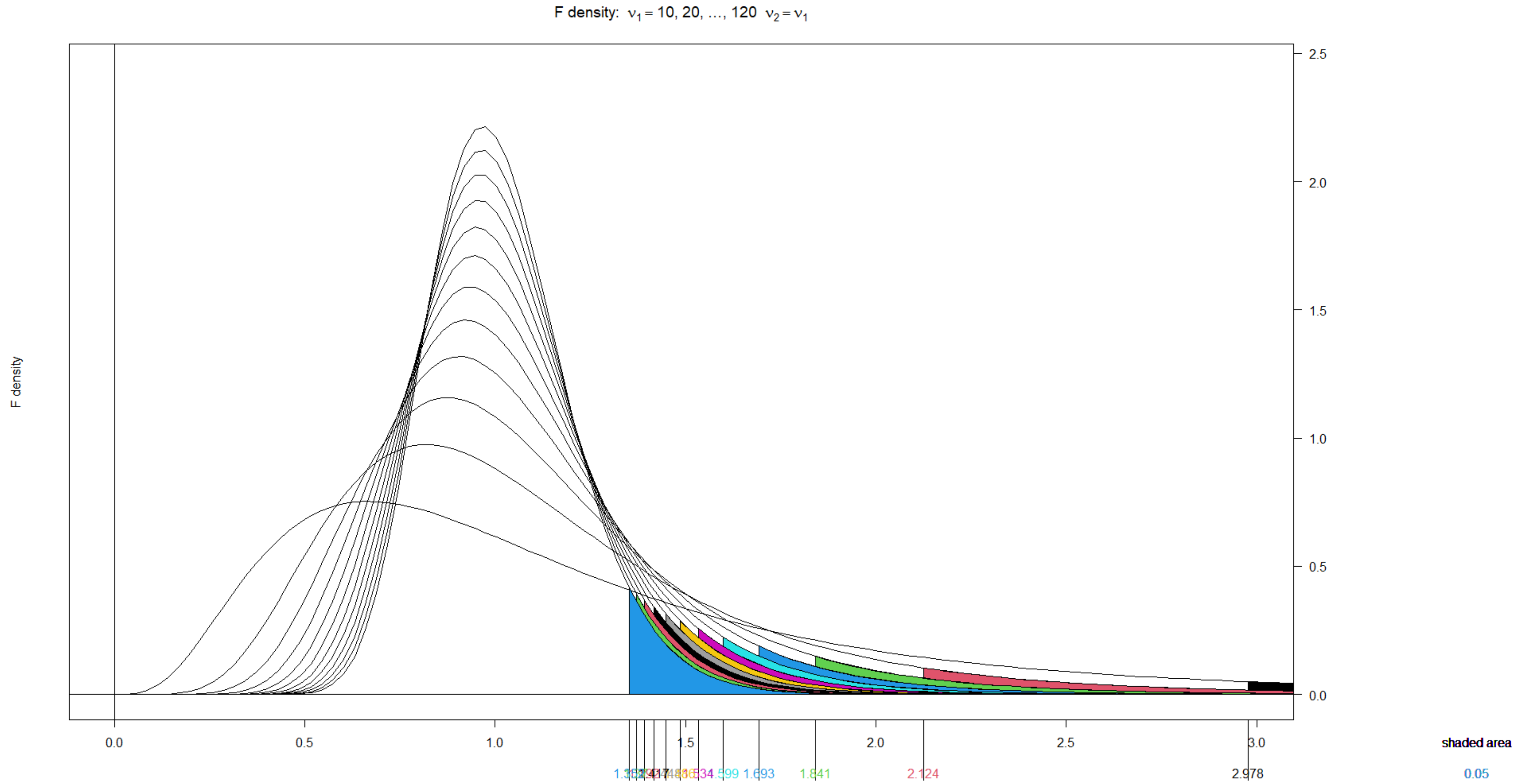
5-5.1 Hypothesis Testing on the Ratio of Two Variances



5-5.1 Hypothesis Testing on the Ratio of Two Variances



5-5.1 Hypothesis Testing on the Ratio of Two Variances



5-5.1 Hypothesis Testing on the Ratio of Two Variances

- **Upper-Tail Percentage Points of the F-Distribution**: The value $f_{\alpha,u,\upsilon}$ is the point on the F-distribution with u numerator and υ denominator degrees of freedom such that the probability of the random variable exceeding this value is α.

- **Mathematical Illustration**: The probability $P(F > f_{\alpha,u,\upsilon}) = \alpha$ corresponds to the area under the curve to the right of the critical value, as shown in **Fig. 5-5**.

- **Data Source**: These specific percentage points are cataloged in **Table IV** of **Appendix A**.

- **Example Calculation**: For $u = 5$ and $v = 10$, **Table IV** of **Appendix A** indicates that $f_{0.05,5,10} = 3.33$, meaning the upper 5% point of the distribution is 3.33.

- **Table Limitations**: Table IV is restricted to providing only upper-tail percentage points ($f_{\alpha,u,v}$ where $\alpha \leq 0.25$) for the F distribution.

- **Calculating Lower-Tail Points**: To determine a lower-tail percentage point ($f_{1-\alpha,u,v}$), you can use the reciprocal of the corresponding upper-tail point with swapped degrees of freedom:

$$f_{1-\alpha,u,v} = \frac{1}{f_{\alpha,v,u}} \qquad (5-20)$$

- **Example Calculation**: To find $f_{0.95,5,10}$, apply equation **(5-20)** by taking the reciprocal of $f_{0.05,10,5}$. Given that $f_{0.05,10,5} = 4.74$, the result is $1/4.74 = 0.211$.

- **The Test Procedure**: This statistical framework establishes the foundation for comparing and testing the equality of variances between two distinct populations.

- **Sample Assumptions**: The method requires two independent random samples (X_{1j} and X_{2j}) drawn from normal populations with unknown means (μ_1, μ_2) and unknown variances (σ_1^2, σ_2^2).

- **Mathematical Ratio**: Using the sample variances (S_1^2 and S_2^2), a ratio is constructed comparing the scaled sample variances to their respective population variances:

$$F = \frac{S_1^2/\sigma_1^2}{S_2^2/\sigma_2^2}$$

- **Distribution Properties**: This specific ratio follows an F-distribution characterized by $n_1 - 1$ degrees of freedom in the numerator and $n_2 - 1$ degrees of freedom in the denominator.

- **Mathematical Foundation**: The underlying logic relies on the fact that $(n_1 - 1)S_1^2/\sigma_1^2$ and $(n_2 - 1)S_2^2/\sigma_2^2$ are independent chi-square random variables with $n_1 - 1$ and $n_2 - 1$ degrees of freedom, respectively.

- **Simplification Under H_0**: When the null hypothesis H_0: $\sigma_1^2 = \sigma_2^2$ is assumed to be true, the population variances cancel out of the equation.

- **Test Statistic**: This cancellation simplifies the ratio to $F_0 = S_1^2/S_2^2$, which follows a standard F_{n_1-1,n_2-1} distribution and serves as the operational basis for the variance test.

- **Summary**: This section provides a complete reference for **Testing Hypotheses on the Equality of Variances of Two Normal Distributions** under the null hypothesis H_0: $\sigma_1^2 = \sigma_2^2$.

- **Test Statistic**: The calculation is based on the ratio of the two sample variances:

$$f_0 = \frac{s_1^2}{s_2^2} \qquad (5-21)$$

- **Testing Parameters**: The framework maps out the **Alternative Hypothesis**, **P-Value**, and **Rejection Criterion for Fixed-Level Tests** across three distinct structural scenarios:
  - **Two-Sided Condition (1)**:
  - **Alternative Hypothesis**: $H_1: \sigma_1^2 \neq \sigma_2^2$
  - **P-Value**: $2 \cdot \min\{1 - F_{F_{n_1-1,n_2-1}}(f_0), F_{F_{n_1-1,n_2-1}}(f_0)\}$
  - **Rejection Criterion for Fixed-Level Tests**: $f_0 > f_{\alpha/2,n_1-1,n_2-1}$ or $f_0 < f_{1-\alpha/2,n_1-1,n_2-1}$

- Upper-Tailed Condition (2):
  - Alternative Hypothesis: $H_1: \sigma_1^2 > \sigma_2^2$
  - P-Value: $1 - F_{F_{n_1-1,n_2-1}}(f_0)$
  - Rejection Criterion for Fixed-Level Tests: $f_0 > f_{\alpha,n_1-1,n_2-1}$

- Lower-Tailed Condition (3):
  - **Alternative Hypothesis**: $H_1: \sigma_1^2 < \sigma_2^2$
  - **P-Value**: $F_{F_{n_1-1,n_2-1}}(f_0)$
  - **Rejection Criterion for Fixed-Level Tests**: $f_0 < f_{1-\alpha,n_1-1,n_2-1}$

- **Visual Map**: The operational bounds for these rejection zones are explicitly detailed in **Fig. 5-6**.

- **EXAMPLE 5-10 Etching Wafers**: This study examines whether two gas mixtures differ in the variability of etched oxide layers on semiconductor wafers, using a sample of 16 wafers per gas and sample standard deviations of $s_1 = 1.96$ and $s_2 = 2.13$ angstroms.

Solution.

- **1. Parameter of interest:** The ratio of the variances of oxide thickness, σ_1^2/σ_2^2, assuming oxide thickness is a normal random variable for both mixtures.

- **2. Null hypothesis, H₀** : $\sigma_1^2 = \sigma_2^2$.

- **3. Alternative hypothesis, H₁** : $\sigma_1^2 \neq \sigma_2^2$.
- **4. Test statistic**: The test statistic equation is $f_0 = s_1^2/s_2^2$.
- **5. Reject H₀ if**: With $n_1 = n_2 = 16$ and $\alpha = 0.05$, reject if $f_0 > f_{0.025,15,15} = 2.86$ or if $f_0 < f_{0.975,15,15} = 0.35$.
- **6. Computations**: Given $s_1^2 = 3.84$ and $s_2^2 = 4.54$, the test statistic is calculated as $f_0 = 3.84/4.54 = 0.85$.
- **7. Conclusions**: Because $0.35 < 0.85 < 2.86$, we cannot reject the null hypothesis at the 0.05 level of significance.

- **Practical engineering conclusion:** There is no strong evidence indicating that either gas results in lower variability, allowing selection based on cost or ease of use.

- **Calculation Approach**: For this two-sided test (H_1: $\sigma_1^2 \neq \sigma_2^2$), since the test statistic $f_0 = 0.8454$ is less than 1, it falls on the lower tail. The cumulative probability of the lower tail is multiplied by 2.

- **Exact p-value**: **0.7481** (or 74.81%)

- **Statistical Interpretation**: Because the p-value (0.7481) is much larger than the significance level ($\alpha = 0.05$), we formally fail to reject the null hypothesis. There is a 74.81% chance of observing a variance ratio this extreme or more due to random sampling noise alone.

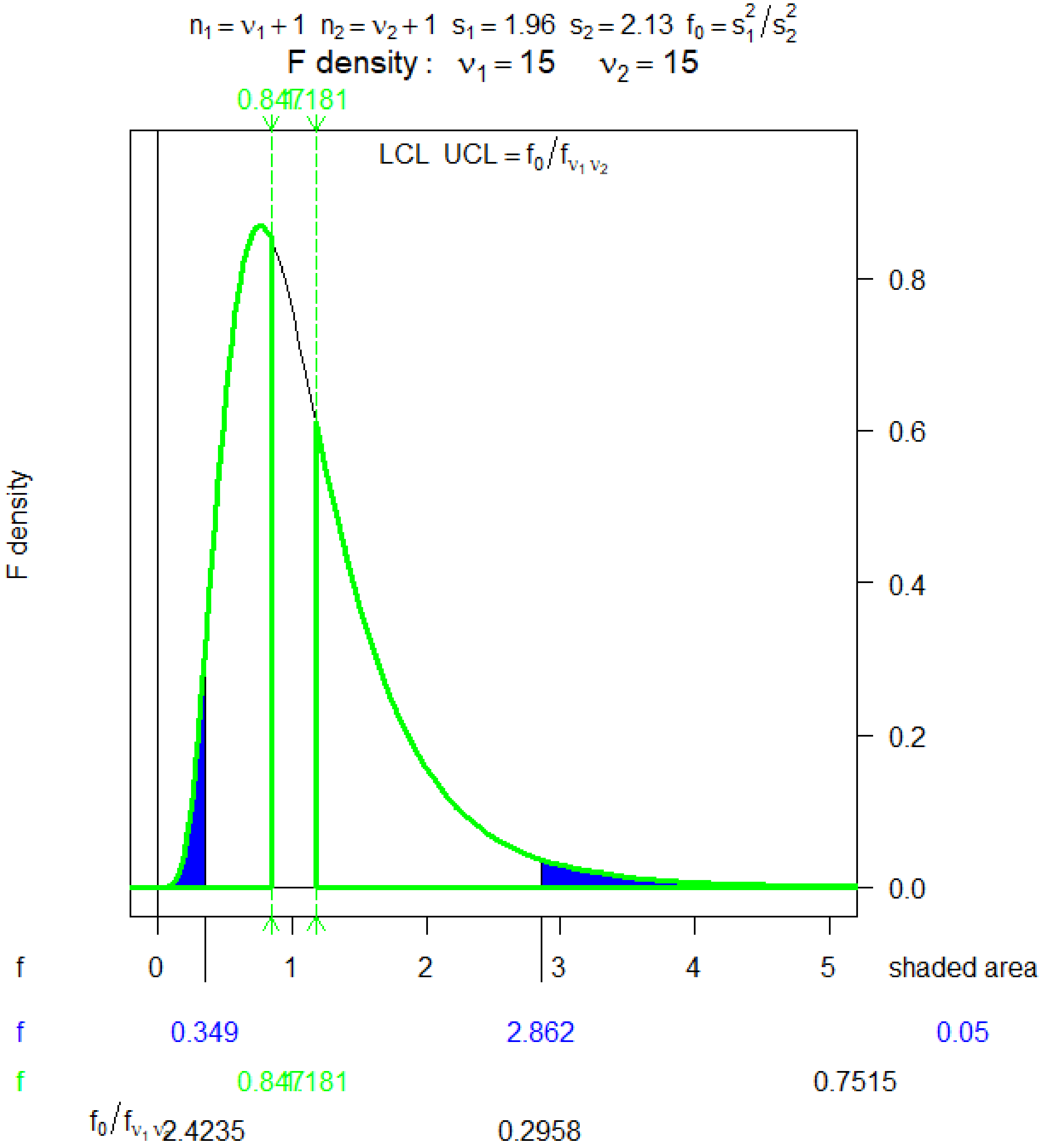
- **Formula Structure**: The $100(1-\alpha)\%$ confidence interval for the ratio of two independent normal variances is defined as:

$$\frac{s_1^2}{s_2^2} f_{1-\alpha/2, n_2-1, n_1-1} \leq \frac{\sigma_1^2}{\sigma_2^2} \leq \frac{s_1^2}{s_2^2} f_{\alpha/2, n_2-1, n_1-1}$$

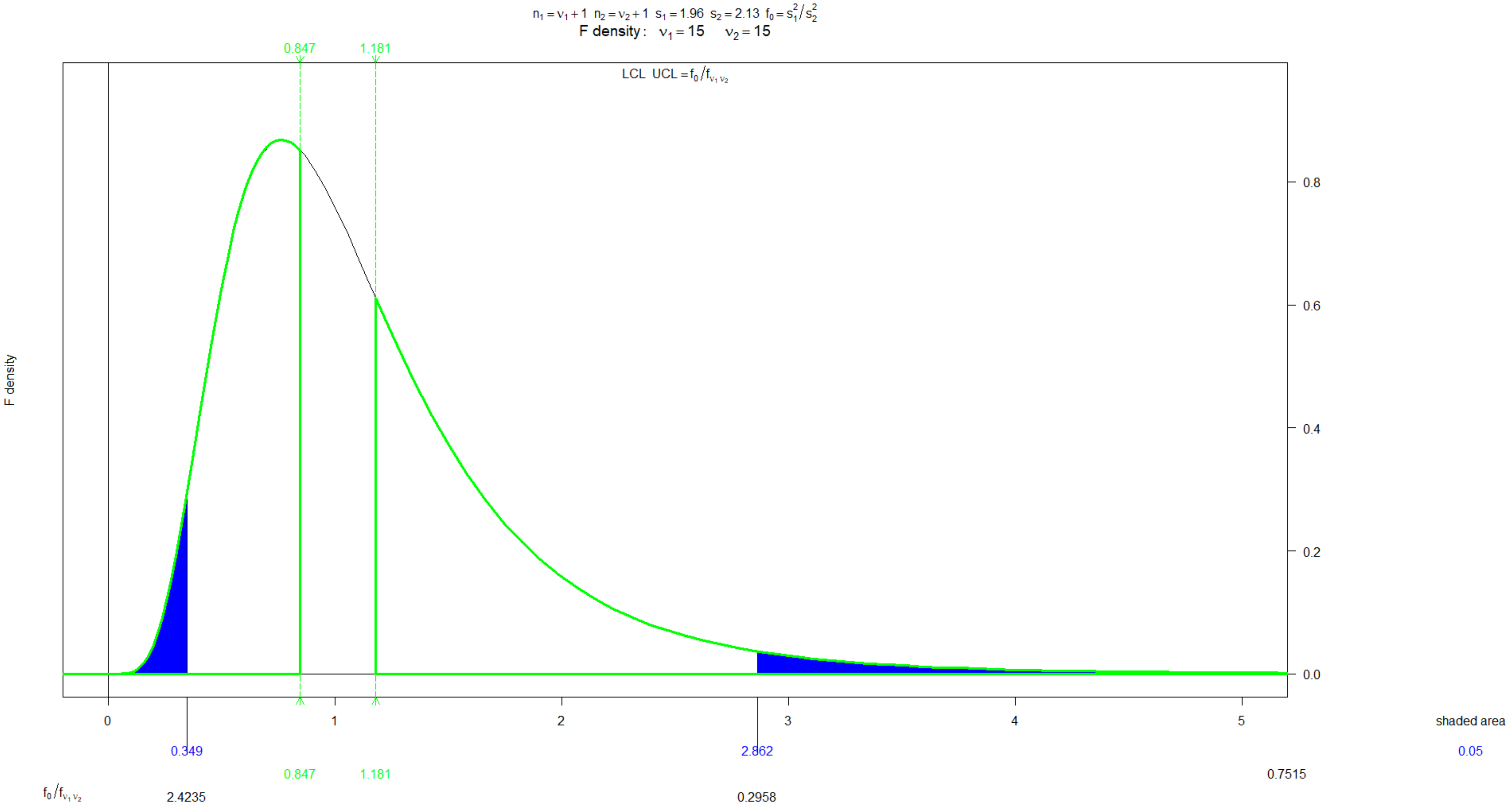
- **Critical Values**: For $\alpha = 0.05$ and $df_1 = 15$, $df_2 = 15$:
  - Upper critical value: $f_{0.025,15,15} = 2.86$
  - Lower critical value (using the reciprocal property): $f_{0.975,15,15} = 1/2.86 = 0.35$

- **Interval Calculation**: Substituting the sample variance ratio $s_1^2/s_2^2 = 0.85$:
  - Lower Bound: $0.85 \times 0.35 = 0.30$
  - Upper Bound: $0.85 \times 2.86 = 2.43$
  - **95% Confidence Interval**: $[0.30, 2.43]$

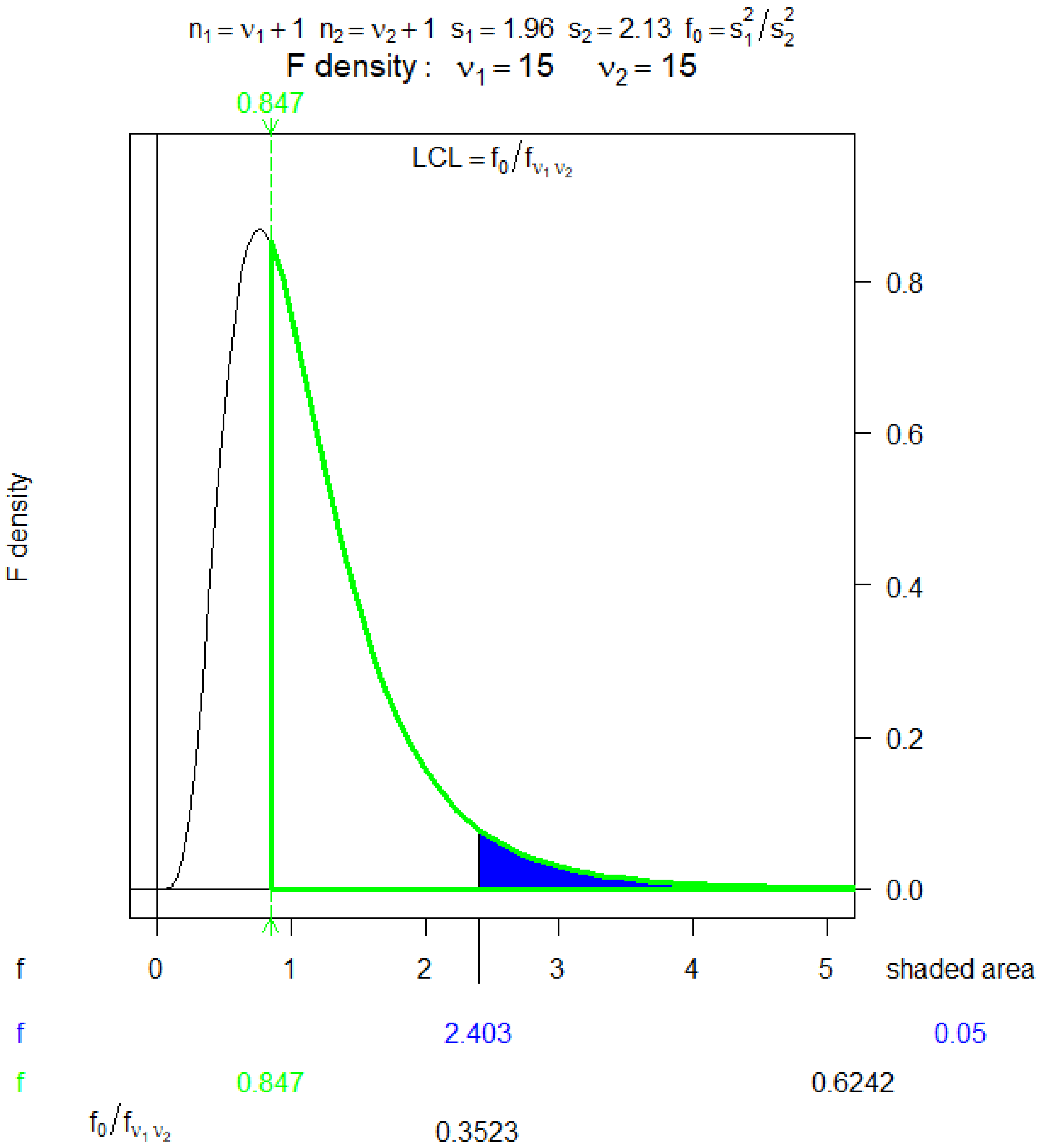
Engineering Interpretation: Because this confidence interval **contains the value 1** (which represents $\sigma_1^2 = \sigma_2^2$), it is fully consistent with our hypothesis test conclusion. We cannot claim that either gas mixture yields a different variance in oxide thickness.



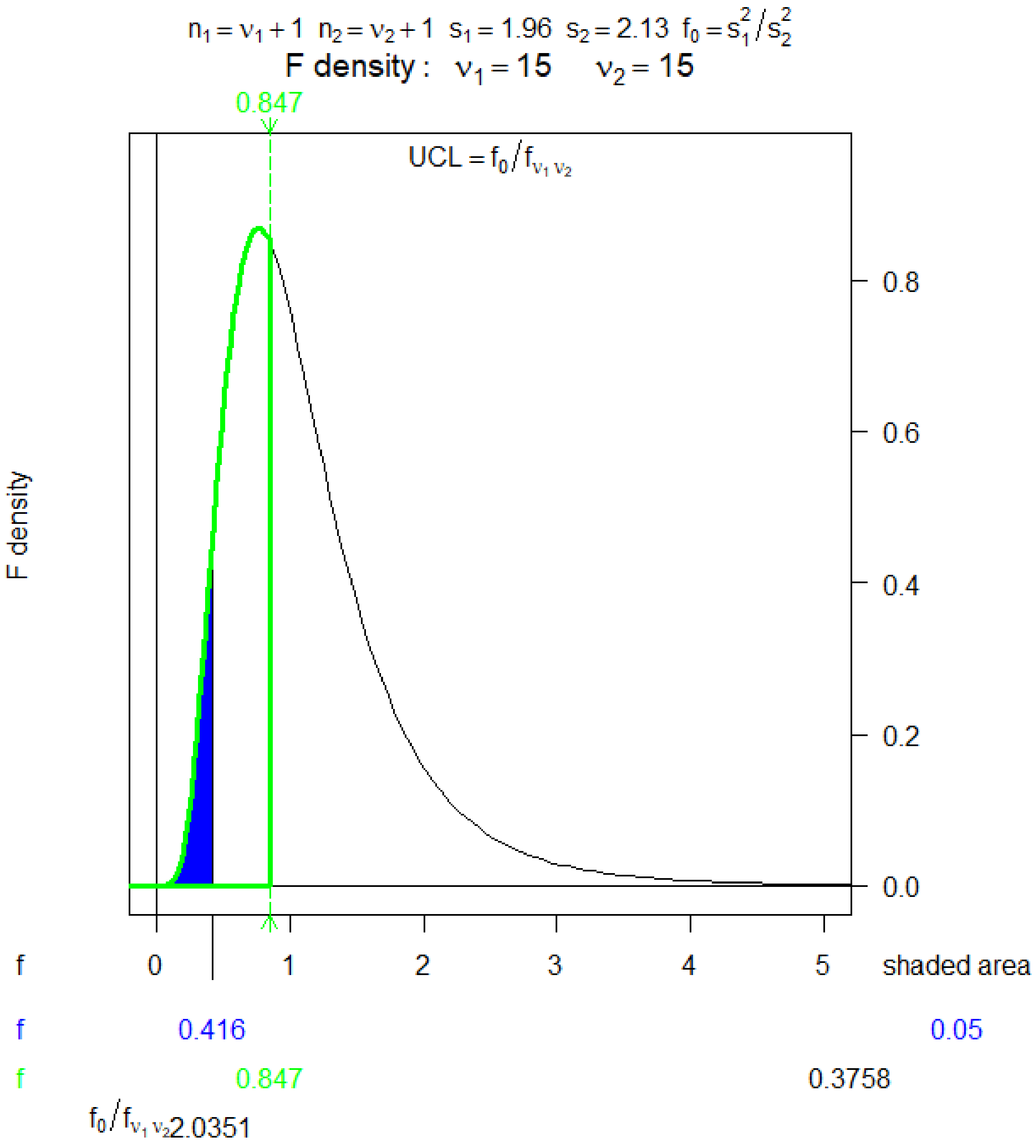
5-5.1 Hypothesis Testing on the Ratio of Two Variances



5-5.1 Hypothesis Testing on the Ratio of Two Variances



5-5.1 Hypothesis Testing on the Ratio of Two Variances



5-5.1 Hypothesis Testing on the Ratio of Two Variances

5-5.2 Confidence Interval on the Ratio of Two Variances

- **Definition**: This statistical method establishes a **Confidence Interval on the Ratio of Variances of Two Normal Distributions** when the true population variances are unknown.

- **Requirements**: The interval requires sample variances (s_1^2, s_2^2) and sample sizes (n_1, n_2) from two independent normal populations.

- **Confidence Interval Formula**: The $100(1-\alpha)\%$ confidence interval for the ratio σ_1^2/σ_2^2 is:

$$(s_1^2/s_2^2)\cdot f_{1-\alpha/2,n_2-1,n_1-1} \le \sigma_1^2/\sigma_2^2 \le (s_1^2/s_2^2)\cdot f_{\alpha/2,n_2-1,n_1-1} \qquad (\mathbf{5-22})$$

- **Critical Values in (5-22)**: The terms $f_{\alpha/2, n_2-1, n_1-1}$ and $f_{1-\alpha/2, n_2-1, n_1-1}$ represent the upper and lower percentage points of the F distribution, calculated using $n_2 - 1$ numerator and $n_1 - 1$ denominator degrees of freedom.

- **EXAMPLE 5-11 Surface Finish**: An engineer is comparing two grinding processes for titanium alloy jet-turbine impellers to identify the one with the least variability in surface roughness. The first process yielded a standard deviation of $s_1 = 5.1$ $(n_1 = 11)$, while the second yielded $s_2 = 4.7$ $(n_2 = 16)$.

Solution.

- **Assumptions**: The analysis assumes that the two processes are independent and that the surface roughness measurements follow a normal distribution.

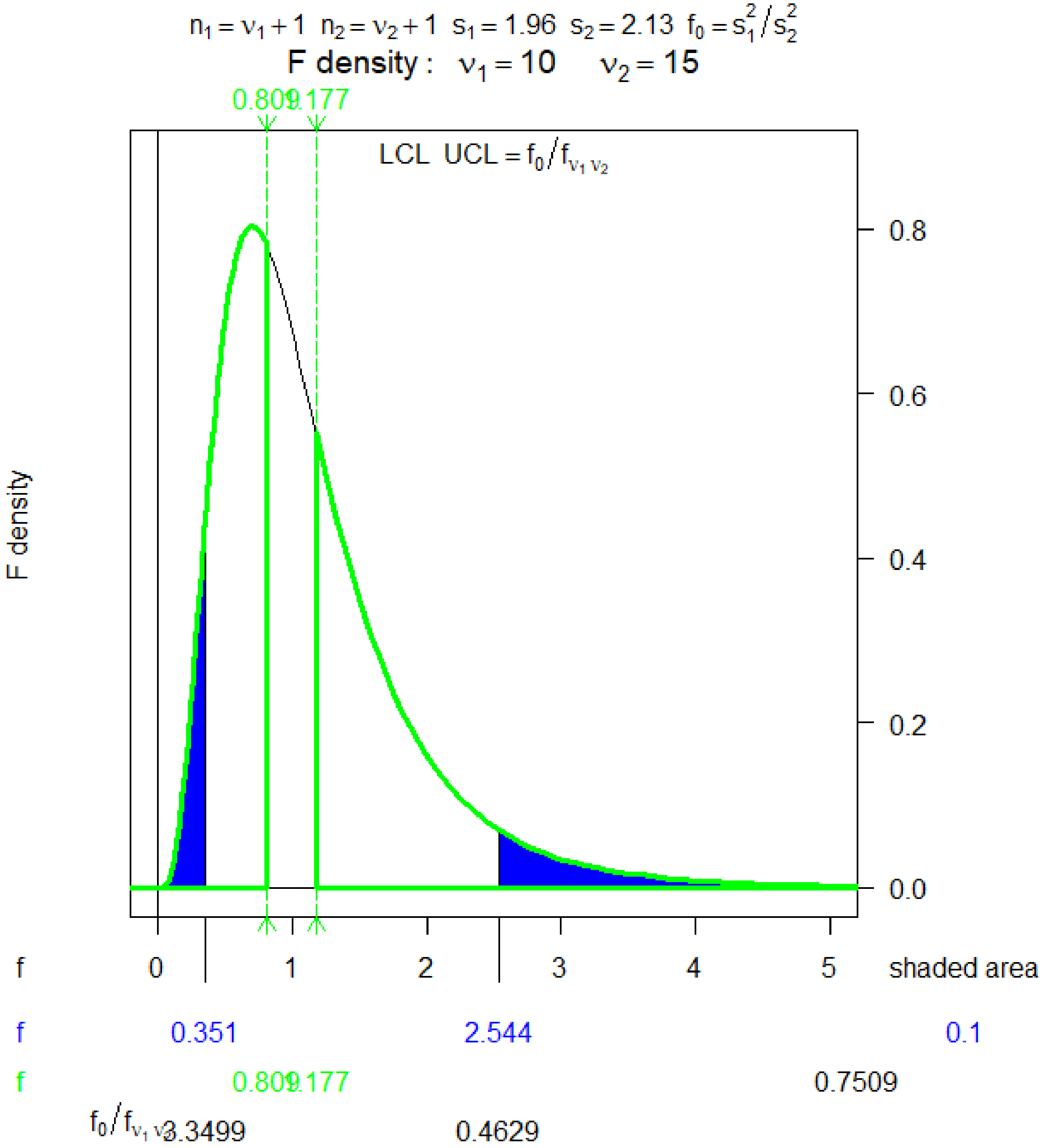
- **Critical Values**: To find the lower bound, the reciprocal identity was used: $f_{0.95,15,10} = 1/f_{0.05,10,15} = 1/2.54 = 0.39$. The upper bound critical value is $f_{0.05,15,10} = 2.85$.

- **Confidence Interval Calculation**: Plugging the values into the formula for a 90% confidence interval on the ratio of variances σ_1^2/σ_2^2:

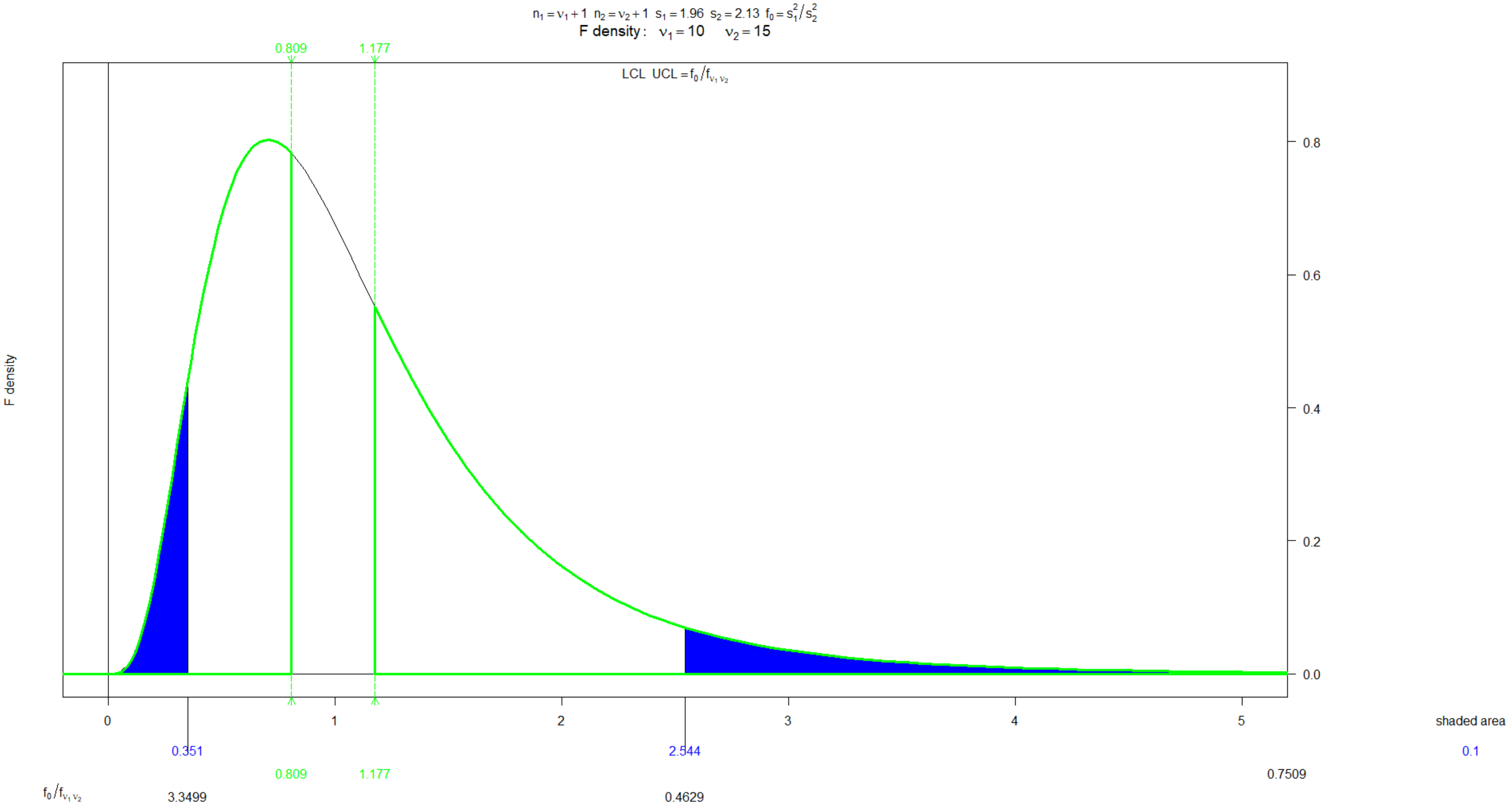
$$(5.1^2/4.7^2) \cdot 0.39 \le \sigma_1^2/\sigma_2^2 \le (5.1^2/4.7^2) \cdot 2.85$$

$$0.46 \le \sigma_1^2/\sigma_2^2 \le 3.35$$

- **Conclusion**: Since the interval includes **unity (1.0)**, there is insufficient evidence at the 90% confidence level to claim that the standard deviations of the two processes are different.



$n_1 = \nu_1 + 1 \quad n_2 = \nu_2 + 1 \quad s_1 = 1.96 \quad s_2 = 2.13 \quad f_0 = s_1^2 / s_2^2$
F density : $\nu_1 = 10 \quad \nu_2 = 15$
0.809
1.177
LCL UCL = $f_0 / f_{\nu_1 \nu_2}$
F density
0.8
0.6
0.4
0.2
0.0
f
0
1
2
3
4
5
shaded area
f
0.351
2.544
0.1
f
0.809
1.177
0.7509
$f_0 / f_{\nu_1 \nu_2}$
3.3499
0.4629



5-5.2 Confidence Interval on the Ratio of Two Variances

One-Sided Confidence Bounds

- **Lower Confidence Bound**: To calculate a $100(1-\alpha)\%$ lower bound for σ_1^2/σ_2^2, the term $f_{1-\alpha/2,n_2-1,n_1-1}$ in the lower limit of equation 5-22 is replaced with $f_{1-\alpha,n_2-1,n_1-1}$, and the upper bound is set to infinity.
- **Upper Confidence Bound**: To find a $100(1-\alpha)\%$ upper bound on σ_1^2/σ_2^2, replace $f_{\alpha/2,n_2-1,n_1-1}$ with f_{α,n_2-1,n_1-1} in the upper limit of equation 5-22, and set the lower bound to 0.
- **Standard Deviation Bounds**: To convert these into confidence bounds for the ratio of standard deviations (σ_1/σ_2), simply take the square root of the interval endpoints or bounds calculated for the variances.